

System Identification III: The Eigensystem Realization Algorithm (ERA)

From an impulse response to a state-space model (A,B,C) .

`Intro.m` and `OKID.m` produce Markov parameters – the impulse response. ERA (Juang & Pappa) turns that sequence into an actual state-space realization, and along the way **reveals the model order** from the rank of a Hankel matrix. You will:

- build the block Hankel matrices $H(0)$ and $H(1)$ from Markov parameters,
- read the order off the singular values, and
- recover (A, B, C) and confirm its poles and response match the truth.

Run with `publish('ERA.m')`, or step through with **Ctrl+Enter**.

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Markov parameters to work from

Take a known 3rd-order discrete system and list its impulse response $Y_k = CA^{k-1}B$. (In practice these come from `OKID`.)

```
A_true = [0.5 0.6 0; -0.3 0.4 0.2; 0 -0.1 0.7];
B_true = [1; 0; 0.5];
C_true = [1 0 -1];
nt = size(A_true,1);

L = 40;
Y = zeros(1,L);
for k = 1:L
    Y(k) = C_true*A_true^(k-1)*B_true;
end
```

Build the block Hankel matrices

With $H(0)_{ij} = Y_{i+j-1}$ and $H(1)_{ij} = Y_{i+j}$ (a one-step shift):

```
s = 15; % Hankel size (>= system order)
H0 = hankel(Y(1:s), Y(s+1:2*s));
H1 = hankel(Y(2:s+1), Y(s+2:2*s+1));
```

The order is the rank of $H(0)$

Notice that only a few singular values are meaningfully nonzero – their count is the model order n . Everything below the "elbow" is numerical dust.

```
sv = svd(H0);
figure
semilogy(sv,'o-','LineWidth',1.2)
grid on
title('Hankel Singular Values Reveal the Order','Interpreter','latex','FontSize',15)
```

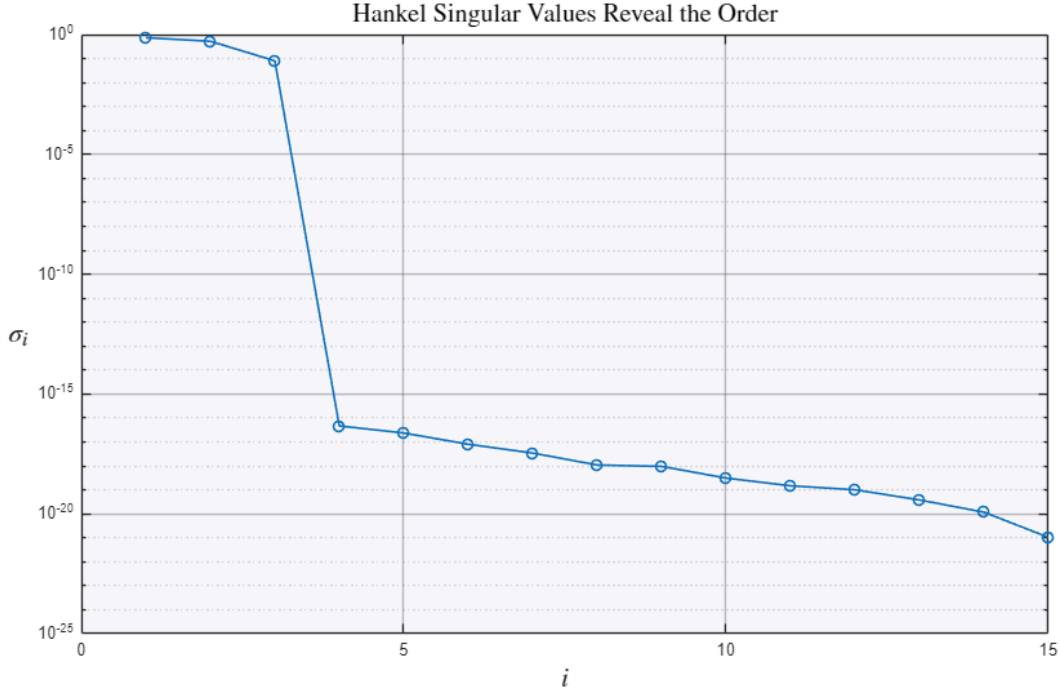
```

ylabel('\sigma_i','Interpreter','latex','FontSize',16); set(get(gca,'YLabel'),'Rotation',0)
xlabel('$i$','Interpreter','latex','FontSize',16)

n = sum(sv > 1e-8*sv(1));          % numerical rank
fprintf('Detected model order n = %d (true = %d)\n', n, nt)

```

Detected model order n = 3 (true = 3)



Recover (A, B, C)

Truncate the SVD $H(0) = R\Sigma V^T$ to order n and apply the ERA formulas:

$$\hat{A} = \Sigma_n^{-1/2} R_n^T H(1) V_n \Sigma_n^{-1/2}, \quad \hat{B} = \Sigma_n^{1/2} V_n^T e_1, \quad \hat{C} = e_1^T R_n \Sigma_n^{1/2}.$$

```

[R,S,V] = svd(H0);
Rn = R(:,1:n); Sn = S(1:n,1:n); Vn = V(:,1:n);
Sh = Sn^(1/2); Si = Sn^(-1/2);
Ah = Si*Rn.'*H1*Vn*Si;
Bh = Sh*Vn(1,:).';
Ch = Rn(1,:)*Sh;

```

Check: same poles, same response

The realization is **not** unique (any similarity transform gives the same input/output behavior), so we compare the invariants – the **poles** and the **impulse response** – not the matrices themselves.

```

fprintf('True poles:      %s\n', mat2str(sort(eig(A_true)).',3))
fprintf('Recovered poles: %s\n', mat2str(sort(eig(Ah)).',3))

Yhat = zeros(1,L);
for k = 1:L
    Yhat(k) = Ch*Ah^(k-1)*Bh;
end
figure

```

```

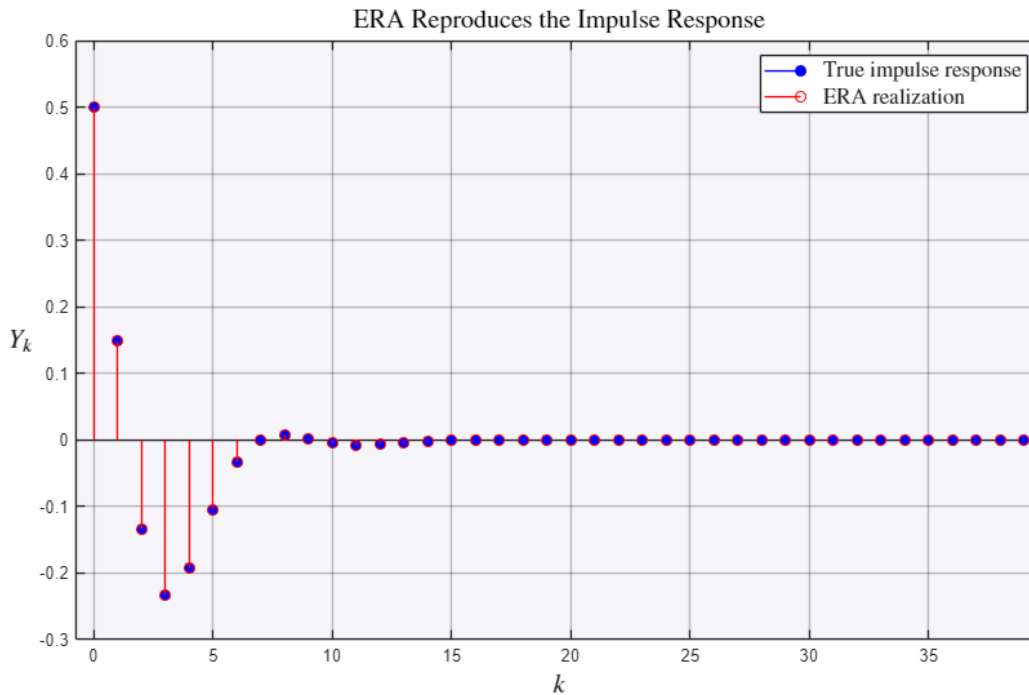
stem(0:L-1, Y, 'b', 'filled')
hold on
stem(0:L-1, Yhat, 'r')
hold off
grid on
legend('True impulse response', 'ERA realization', 'Interpreter', 'latex', 'FontSize', 13)
title('ERA Reproduces the Impulse Response', 'Interpreter', 'latex', 'FontSize', 15)
ylabel('$Y_k$', 'Interpreter', 'latex', 'FontSize', 16); set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$k$', 'Interpreter', 'latex', 'FontSize', 16)
fprintf('Impulse-response reconstruction error: %.2e\n', norm(Yhat-Y)/norm(Y))

```

```

True poles:      [0.458-0.44i 0.458+0.44i 0.684]
Recovered poles: [0.458-0.44i 0.458+0.44i 0.684]
Impulse-response reconstruction error: 1.64e-15

```



Try it yourself

- Add noise ($Y = Y + 1e-3 \cdot \text{randn}(1, L);$) and watch the singular values stop dropping sharply – the elbow blurs and choosing n becomes a judgement call.
- Force $n = 2$ and see ERA return the best 2nd-order approximation of this 3rd-order system.

The full pipeline: OKID (OKID.m) delivers Markov parameters even for ringing systems; ERA turns them into (A, B, C) . Together they take you from raw data to a usable state-space model.

Summary

- ERA builds Hankel matrices from Markov parameters; their rank is the model order.
- A truncated SVD yields a state-space realization (A, B, C) .
- Poles and impulse response are recovered (up to noise); the internal state coordinates are arbitrary.